

5(a)(i)	phagocytosis (of, dead cells / pathogens / AW) ; A endocytosis of pathogens	1
5(a)(ii)	(sodium chloride) solution has lower water potential (than cytoplasm) ; ora A Ψ for water potential water, leaves / AW, cell by <u>osmosis</u> ;	2
5(b)(i)	13.56 ; A 13.6 / 14	1

Question	Answer	Marks
5(b)(ii)	<i>allow (red) blood cell count for mean red blood cell count</i> <i>allow ora</i> <i>any four from:</i> - description of data ; e.g. lowest mean red blood cell count for, (population) A / lowest altitude (populations) B, C and D / all other altitudes, have a higher mean red blood cell count than A / 695m altitude (population) B has highest mean red blood cell count (population) B has higher mean red blood cell count than C <u>and</u> D <i>physiological explanation population A compared to B/C/D</i> - decrease in partial pressure of oxygen in the atmosphere with increase in altitude / AW or lower partial pressure of / less, oxygen in, inhaled air / lungs / alveoli AW <i>ref. to haemoglobin ;</i> e.g. lower saturation of haemoglobin with oxygen (in lungs) lower affinity of haemoglobin for oxygen (in lungs) more haemoglobin required to carry sufficient oxygen to tissues - suggestion of increase in count at 1676 m and above ; e.g. rate of production of red blood cells increases higher erythropoietin (EPO) levels <i>ref. to changed proportion of plasma to red blood cells / plasma volume decreases</i> - little difference between B, C and D / AW ; - differences in altitude are not great / AW ; <i>in context of B, C, D</i> <i>ref. to genetic differences / adaptation to environment ;</i> <i>allow if general statement includes population A</i> - AVP ; e.g. <i>ref. to differences in demographics</i>	4

Question	Answer	Marks
5(c)	less haemoglobinic acid is formed / fewer hydrogen ions to bind to haemoglobin (to cause dissociation of oxygen) ; <i>any one from:</i> binds to allosteric site / site other than active site (of enzyme), qualified ; e.g. causes change in shape of active site substrate cannot bind to active site fewer / no, enzyme-substrate / ES, complex formation prevents / reduces / AW, carbonic acid / H_2CO_3, formation / AW ; A fewer hydrogen ions produced from carbonic acid dissociation	2

Question	Answer	Marks
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